

The exact arithmetic coefficient and theta-generating join

13 September 2026

1 Source, attribution, and original coordinates

This chapter proves the coefficient and cochain maps needed to integrate *Tau Gamma Convolution Descent*, whose original `NOTE.tex` has SHA256

7cf984d87de405b815250b44454c4ee27c8ab46574da245b377ebc6e49327fa2.

The analytic generating identities (GC.24)–(GC.26) were supplied by the *Zeta Function Literature & Local Corpus Foundation* task on 13 September 2026. The proofs here verify and integrate that supplied join; they do not claim its discovery. The original gamma generating integral and monic family are the existing TG construction, with the classical Meixner–Pollaczek parameters retained.

All scalar source coordinates remain

$$s = \frac{1}{2} + it, \quad S = s_1 + \cdots + s_k = \frac{k}{2} + iu, \quad u = t_1 + \cdots + t_k. \quad (\text{GC.1})$$

Here $k \geq 1$ is an integer. We use inner products conjugate-linear in the first argument. For a polynomial, an overline over its name means coefficientwise conjugation. In particular $\overline{P(c + iu)} = \overline{P}(c - iu)$ when c, u are real.

1.1 The analytic source and its actual theta boundary

Let $D = -x\partial_x$. Let V be the complex vector space of even Schwartz functions ϕ on $\mathbb{R}$ with $\phi(0) = 0$ and $\int_{\mathbb{R}} \phi(x) dx = 0$. Let $\mathcal{B}$ be the smooth functions on $\mathbb{R}_{>0}$ such that for every $a > 0$ and every integer $j \geq 0$,

$$D^j F(x) = O(x^a) \quad (x \downarrow 0), \quad D^j F(x) = O(x^{-a}) \quad (x \rightarrow \infty).$$

The original theta map and cochain grading are

$$\Theta : V \longrightarrow \mathcal{B}, \quad \Theta \phi(x) = \sum_{n \neq 0} \phi(nx), \quad [V \xrightarrow{\Theta} \mathcal{B}] \text{ in degrees } 0, 1. \quad (\text{GC.2})$$

The sum and its differentiated sums converge at infinity by the Schwartz bounds. Poisson summation gives its corresponding bounds at zero: the zero Fourier coefficient vanishes by $\int \phi = 0$, and the original zero term vanishes by $\phi(0) = 0$. Differentiation preserves both conditions, because $D\phi(0) = 0$ and integration by parts gives $\int D\phi = \int \phi = 0$. Thus $D\Theta = \Theta D$ on these spaces. The rapid bounds make the Mellin integral

$$\mathcal{M}F(s) = \int_0^\infty F(x)x^{s-1} dx$$

entire for $F \in \mathcal{B}$ and give $\mathcal{M}(DF)(s) = s\mathcal{M}F(s)$, including the vanished endpoint terms.

For clarity the seed and all its constants can be constructed directly. Put

$$\phi_0(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad \vartheta(z) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 z^2}, \quad f_0 = D(D-1)\vartheta = \Theta\phi_0. \quad (\text{GC.3})$$

Gaussian integration gives $\int 4\pi^2 x^4 e^{-\pi x^2} dx = 3$ and $\int 6\pi x^2 e^{-\pi x^2} dx = 3$ on $\mathbb{R}$, and $\phi_0(0) = 0$, so $\phi_0 \in V$. The Gaussian Fourier transform in the convention $\widehat{f}(y) = \int f(x)e^{-2\pi i xy} dx$ is $e^{-\pi y^2}$: differentiation under the integral and integration by parts give $\widehat{f}' = -2\pi y \widehat{f}$, and the integral at zero is one. Periodizing this Gaussian gives an absolutely convergent Fourier series with those computed coefficients. Evaluating it at zero proves $\vartheta(1/x) = x\vartheta(x)$ for $x > 0$. Holomorphic continuation gives the same identity for $|\arg z| < \pi/4$, where both theta series converge normally on compact subsets.

For $RF(x) = x^{-1}F(1/x)$, direct differentiation gives $DR = R(1-D)$. Hence $D(D-1)$ commutes with R and

$$f_0(1/z) = zf_0(z). \quad (\text{GC.4})$$

On every closed narrower sector, each derivative of f_0 at infinity is bounded by a polynomial times $e^{-c|z|^2}$ for some $c > 0$. Equation (GC.4), differentiated finitely many times, proves rapid decay at zero too. In particular $f_0 \in \mathcal{B}$.

For $\Re s > 1$, absolute convergence permits termwise Mellin integration of $2 \sum_{n \geq 1} \phi_0(nx)$. The change $y = \pi n^2 x^2$ gives

$$\begin{aligned} \mathcal{M}f_0(s) &= \pi^{-s/2} \zeta(s) \{4\Gamma(s/2 + 2) - 6\Gamma(s/2 + 1)\} \\ &= s(s-1)\pi^{-s/2} \Gamma(s/2) \zeta(s) =: g(s) = 2\xi(s). \end{aligned} \quad (\text{GC.5})$$

The first line retains the theta factor two and the integration Jacobian $1/2$. The gamma recurrence proves the second line. The Mellin integral gives its entire continuation.

Let $h(s) = \prod_{\rho} (s - \rho)^{m_{\rho}}$ be a finite monic packet of actual nontrivial zeros of g , each selected with its complete order. The empty packet is $h = 1$. Write $d = \deg h$ and $v_h = g/h$. At each selected centre, cancelling its exact local factor constructs the analytic value of v_h ; hence v_h is entire. It is nonzero at each selected centre and is not identically zero.

We record the bounds that justify the integrals below. For any closed real interval of σ , any $0 < \beta < \pi/4$, and any $j \geq 0$,

$$|v_h(\sigma + it)| \leq C_{j,\beta,h} (1 + |t|)^{-j} e^{-\beta|t|}. \quad (\text{GC.6})$$

To prove this first for g , rotate its Mellin contour to $z = e^{i\beta}x$ for $t \geq 0$ and to $z = e^{-i\beta}x$ for $t < 0$. The sector estimates after (GC.4) make the two closing arcs vanish, uniformly for the stated σ interval. The rotation contributes $e^{\mp\beta t}$ and a unit-modulus factor depending on σ . Repeated integration by parts along that ray gives every displayed power of $(1 + |t|)^{-1}$; all ray endpoint terms vanish by the same estimates. For $|t|$ larger than twice the moduli of all packet centres relative to that σ interval, monicity gives a lower bound for $|h(\sigma + it)|$ by a positive constant times $|t|^d$. On the remaining compact rectangle the cancelled entire quotient is bounded. This proves (GC.6) for v_h .

For any real σ define

$$F_h(x) = \frac{1}{2\pi} \int_{\mathbb{R}} v_h(\sigma + it) x^{-\sigma - it} dt. \quad (\text{GC.7})$$

Horizontal rectangle estimates from (GC.6) prove independence of σ . Differentiating in x multiplies the integrand by a power of $\sigma + it$ and remains absolutely convergent. Choosing σ arbitrarily large positive at infinity and arbitrarily large negative at zero proves $F_h \in \mathcal{B}$. Fourier inversion in $\log x$, with the displayed $1/(2\pi)$ constant, gives $\mathcal{M}F_h = v_h$. Consequently Mellin injectivity gives

$$h(D)F_h = f_0 = \Theta\phi_0. \quad (\text{GC.8})$$

Define the actual density, mass, and Laplace integral by

$$w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad M_h(\theta) = \int_{\mathbb{R}} e^{\theta t} w_h(t) dt. \quad (\text{GC.9})$$

Equation (GC.6) proves holomorphy of M_h on $|\Re \theta| < \pi/2$, with $M_h^{(j)}(\theta) = \int t^j e^{\theta t} w_h(t) dt$ there. Indeed any compact substrip is dominated using a β for which 2β is larger than its real width. The mass is positive because v_h is a nonzero analytic function; its real-line zeros are discrete. Every original spectral moment and its 2π factor are retained.

2 The gamma reference with its actual mass

Fix $\lambda > 0$ and write $\alpha = 2\lambda$. Define

$$d\sigma_\lambda(t) = r_\lambda(t) dt, \quad r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad c_\lambda = 2^{1-2\lambda} \Gamma(2\lambda). \quad (\text{GC.10})$$

The product of two gamma integrals, changed by $(a, b) = (ry, (1-r)y)$ with Jacobian y , proves the beta integral. Its substitution $r/(1-r) = e^x$ gives

$$\int_{\mathbb{R}} (2 \cosh(x/2))^{-2\lambda} e^{ivx} dx = \frac{\Gamma(\lambda + iv) \Gamma(\lambda - iv)}{\Gamma(2\lambda)}.$$

The integrand before Fourier transformation and its second derivative are integrable. Two integrations by parts make its transform integrable, so Fourier inversion is valid. Alternatively, multiplying that transform by $e^{-\varepsilon v^2}$ gives an absolutely integrable Gaussian approximation and proves the same inversion by dominated convergence. The substitution $t = 2v$, with Jacobian two, therefore proves

$$\int_{\mathbb{R}} e^{iyt} r_\lambda(t) dt = c_\lambda (\cosh y)^{-2\lambda}. \quad (\text{GC.11})$$

The beta integrand extends holomorphically to $|\Im x| < \pi$ using the branch positive on $\mathbb{R}$. On every closed narrower strip it is bounded by a constant times $e^{-\lambda|\Re x|}$. Shifting the contour up for $v > 0$ and down for $v < 0$ proves $|\Gamma(\lambda + iv)|^2 \leq C_{\lambda,b} e^{-b|v|}$ for every $b < \pi$. Thus σ_λ has exponential moments of every order less than $\pi/2$ in the original t variable.

Fourier uniqueness for finite measures and multiplication of (GC.11) give the literal sum density

$$r_{\lambda,k} := r_\lambda^{*k} = \frac{c_\lambda^k}{c_{k\lambda}} r_{k\lambda}, \quad \frac{c_\lambda^k}{c_{k\lambda}} = 2^{k-1} \frac{\Gamma(2\lambda)^k}{\Gamma(2k\lambda)}. \quad (\text{GC.12})$$

Its mass is c_λ^k . Analytic continuation justified by the proved exponential moments gives

$$Z_{\lambda,k}^\Gamma(\theta) = \int e^{\theta u} r_{\lambda,k}(u) du = c_\lambda^k (\cos \theta)^{-k\alpha}, \quad |\Re \theta| < \pi/2. \quad (\text{GC.13})$$

The power uses the analytic logarithm of $\cos \theta$ equal to zero at zero; this cosine has no zeros on the stated strip.

For $|z| < 1$, let

$$\begin{aligned} \mathcal{G}_\lambda(z, t) &= (1 - iz)^{-\lambda + it/2} (1 + iz)^{-\lambda - it/2} \\ &= \sum_{n \geq 0} \frac{b_n^{(\lambda)}(t)}{n!} z^n. \end{aligned} \quad (\text{GC.14})$$

Both logarithms are the principal logarithms and vanish at zero, since $1 \pm iz$ have positive real part. For real z, w near zero, the Laplace transform and the elementary cosine addition identity give

$$\int \mathcal{G}_\lambda(z, t) \mathcal{G}_\lambda(w, t) d\sigma_\lambda(t) = c_\lambda (1 - zw)^{-\alpha}.$$

On a sufficiently small closed bidisc all derivatives of the integrands are bounded by a polynomial times $e^{a|t|}$ with $a < \pi/2$. The exponential moment proof permits their integration. Coefficient comparison then proves, with the monic leading coefficient obtained from $\arctan z = z + O(z^3)$,

$$\begin{aligned} \int b_n^{(\lambda)} b_j^{(\lambda)} d\sigma_\lambda &= c_\lambda n! (\alpha)_n \delta_{nj}, \\ b_0^{(\lambda)} &= 1, \quad b_1^{(\lambda)} = t, \quad b_{n+1}^{(\lambda)} = t b_n^{(\lambda)} - n(n + \alpha - 1) b_{n-1}^{(\lambda)}. \end{aligned} \tag{GC.15}$$

The recurrence follows directly by differentiating (GC.14): $(1+z^2)\partial_z \mathcal{G}_\lambda = (t - \alpha z) \mathcal{G}_\lambda$. The polynomials have real coefficients.

They are complete in $L^2(r_{\lambda,k} du)$ at parameter $k\lambda$. In fact, an orthogonal vector f defines the finite complex measure $f(u) r_{\lambda,k}(u) du$. Cauchy–Schwarz and exponential integrability give its Fourier transform on a nonempty complex strip. All derivatives at zero vanish by polynomial orthogonality, so analytic uniqueness and Fourier uniqueness imply $f = 0$ almost everywhere. This argument uses the full measure mass.

3 The sum projection and exact coefficient powers

On the actual spaces

$$\mathcal{H}_k = L^2(\mathbb{R}^k, \prod_i d\sigma_\lambda(t_i)), \quad \mathcal{H}_\Sigma = L^2(\mathbb{R}, r_{\lambda,k}(u) du),$$

let $U_\Sigma f(\mathbf{t}) = f(\sum_i t_i)$. Fubini and (GC.12) prove that it is an isometry. The affine fibre chart has Jacobian one, so its adjoint for $k \geq 2$ is

$$(C_\Sigma F)(u) = \frac{1}{r_{\lambda,k}(u)} \int_{\mathbb{R}^{k-1}} F(t_1, \dots, t_{k-1}, u - \sum_{i < k} t_i) \prod_{i < k} r_\lambda(t_i) r_\lambda(u - \sum_{i < k} t_i) d^{k-1}t. \tag{GC.16}$$

The denominator is positive everywhere. Fibre Cauchy–Schwarz and Fubini give $\|C_\Sigma F\| \leq \|F\|$, so the formula defines the indicated adjoint on all of $\mathcal{H}_k$. When $k = 1$, both maps are the identity. In every case $C_\Sigma U_\Sigma = 1$ and $P_\Sigma = U_\Sigma C_\Sigma$ is the orthogonal projection. The maps

$$F \mapsto (C_\Sigma F, (1 - P_\Sigma)F), \quad (f, n) \mapsto U_\Sigma f + n, \quad n \in \ker C_\Sigma \tag{GC.17}$$

are inverse Hilbert isometries. This retains the entire relative fibre.

Multiplying the k generating functions gives a finite identity at each coefficient:

$$b_n^{(k\lambda)}(\sum_i t_i) = \sum_{n_1 + \dots + n_k = n} \frac{n!}{\prod_i n_i!} \prod_i b_{n_i}^{(\lambda)}(t_i). \tag{GC.18}$$

Pairing it with a product polynomial and using (GC.15) leaves only its total degree $n = \sum_i n_i$. The pairing at that degree is $c_\lambda^k n! \prod_i (\alpha)_{n_i}$ and the sum norm is $c_\lambda^k n! (k\alpha)_n$. Completeness therefore proves

$$C_\Sigma \prod_i b_{n_i}^{(\lambda)} = \frac{\prod_i (\alpha)_{n_i}}{(k\alpha)_n} b_n^{(k\lambda)}. \tag{GC.19}$$

The squared norm of the retained complementary polynomial is exactly

$$c_\lambda^k \left[\prod_i n_i! (\alpha)_{n_i} - \frac{n! \prod_i (\alpha)_{n_i}^2}{(k\alpha)_n} \right], \quad (\text{GC.20})$$

by the orthogonal decomposition (GC.17).

For the original arithmetic density define, for every $\lambda > 0$,

$$\begin{aligned} A_{h,\lambda}(t) &= \frac{v_h(1/2 + it)}{\Gamma(\lambda + it/2)}, \quad B_{h,\lambda} = |A_{h,\lambda}|^2, \quad w_h = B_{h,\lambda} r_\lambda, \\ c_{h,n} &= \frac{\int b_n^{(\lambda)}(t) w_h(t) dt}{c_\lambda n! (\alpha)_n}, \quad \mathcal{A}_h(z) = \sum_{n \geq 0} (\alpha)_n c_{h,n} z^n, \\ d_{h,k,n} &= [z^n] \mathcal{A}_h(z)^k. \end{aligned} \quad (\text{GC.21})$$

The notation $\mathcal{A}_h$ retains the fixed parameter λ . The complex amplitude $A_{h,\lambda}$ and the generating function $\mathcal{A}_h$ are different specified maps. The integrals in (GC.21) converge by (GC.6), before any convergence claim about the power series. Each $d_{h,k,n}$ is the finite expression

$$d_{h,k,n} = \sum_{n_1 + \dots + n_k = n} \prod_i (\alpha)_{n_i} c_{h,n_i}. \quad (\text{GC.22})$$

Let $m_{h,k} = w_h^{*k}$. Absolute moments and Fubini give, by (GC.18),

$$\int_{\mathbb{R}} b_n^{(k\lambda)}(u) m_{h,k}(u) du = c_\lambda^k n! d_{h,k,n}. \quad (\text{GC.23})$$

This finite identity holds for every $\lambda > 0$ and every n ; it does not require an infinite product to be interchanged with an integral. In particular $c_{h,0} = \mu_h/c_\lambda$ and the total mass is μ_h^k .

4 The supplied analytic generating join, with its branches

Theorem 4.1 (The exact analytic generating function). *For the original arithmetic measure and every $\lambda > 0$, the series in (GC.21) converges holomorphically on $|z| < 1$ and is exactly*

$$\boxed{\mathcal{A}_h(z) = \frac{1}{c_\lambda} (1 + z^2)^{-\lambda} M_h(\arctan z).} \quad (\text{GC.24})$$

Here $\arctan z = (\log(1 + iz) - \log(1 - iz))/(2i)$, and every logarithm is the branch inherited from zero. For real $|\theta| < \pi/4$,

$$\boxed{M_h(\theta) = c_\lambda \sec^{2\lambda} \theta \mathcal{A}_h(\tan \theta).} \quad (\text{GC.25})$$

With the original tensor mass and reference mass,

$$\boxed{\frac{Z_{h,k}(\theta)}{Z_{\lambda,k}^\Gamma(\theta)} = \mathcal{A}_h(\tan \theta)^k, \quad Z_{h,k}(\theta) = \int e^{\theta u} m_{h,k}(u) du = M_h(\theta)^k.} \quad (\text{GC.26})$$

Proof. For $|z| < 1$, both $1 \pm iz$ have positive real part. Their principal logs are analytic, and their sum is the analytic logarithm of $1 + z^2$ vanishing at zero. Direct substitution in (GC.14) gives

$$\mathcal{G}_\lambda(z, t) = (1 + z^2)^{-\lambda} e^{t \arctan z}. \quad (\text{GC.27})$$

There is no omitted phase in this identity: the $it/2$ coefficient multiplies $\log(1 - iz) - \log(1 + iz) = -2i \arctan z$.

The ratio $w = (1 + iz)/(1 - iz)$ has

$$\Re w = \frac{1 - |z|^2}{|1 - iz|^2} > 0, \quad \tan(2\Re \arctan z) = \frac{2\Re z}{1 - |z|^2}.$$

The argument of w lies in $(-\pi/2, \pi/2)$ and equals $2\Re \arctan z$. Thus for every $0 \leq r < 1$,

$$|z| \leq r \implies |\Re \arctan z| \leq \arctan r < \pi/4, \quad |\mathcal{G}_\lambda(z, t)| \leq (1 - r^2)^{-\lambda} e^{\arctan(r)|t|}. \quad (\text{GC.28})$$

For the angle inequality use monotonicity of arctangent and $2|\Re z|/(1 - |z|^2) \leq 2r/(1 - r^2)$, together with $2\arctan r \in [0, \pi/2)$. For the modulus inequality use $|1 + z^2| \geq 1 - r^2$ and real $\lambda > 0$.

Equation (GC.6) makes the right side of (GC.28) integrable against $w_h(t) dt$, uniformly on every compact subdisc. Dominated integration therefore defines a holomorphic function $c_\lambda^{-1} \int \mathcal{G}_\lambda(z, t) w_h(t) dt$ on the whole unit disc. One explicit justification is Morera's theorem: domination permits exchanging the integral with each triangle contour, and the inner holomorphic contour integral is zero. For its derivatives on $|z| \leq r$, choose $r < R < 1$ and apply Cauchy's derivative formula on circles of radius $R - r$; the kernel derivative $|\partial_z^n \mathcal{G}_\lambda(z, t)|$ is bounded by $n!(R - r)^{-n}(1 - R^2)^{-\lambda} e^{\arctan(R)|t|}$. This integrable bound justifies every derivative interchange.

At zero, (GC.14) now gives

$$\mathcal{A}_h^{(n)}(0) = \frac{1}{c_\lambda} \int b_n^{(\lambda)}(t) w_h(t) dt = n!(\alpha)_n c_{h,n}. \quad (\text{GC.29})$$

Thus the Taylor series of this holomorphic integral is exactly the original coefficient series. Substitution of (GC.27) in the integral proves (GC.24). All its constants occur before the integration.

For real $|\theta| < \pi/4$, $|\tan \theta| < 1$, $\arctan(\tan \theta) = \theta$ on the specified branch, and $1 + \tan^2 \theta = \sec^2 \theta > 0$. Solving (GC.24) for $M_h(\theta)$ gives (GC.25). Finally the absolute integral of $\prod_i e^{\theta t_i} w_h(t_i)$ is finite throughout $|\Re \theta| < \pi/2$. Fubini gives $Z_{h,k} = M_h^k$ with no mass division. Divide the k th power of (GC.25) by (GC.13) to obtain (GC.26). At zero this ratio is explicitly $(\mu_h/c_\lambda)^k$. $\square$

The inverse also has a precise complex domain. If $|\Re \theta| < \pi/4$, then

$$|\tan \theta|^2 = \frac{\cosh(2\Im \theta) - \cos(2\Re \theta)}{\cosh(2\Im \theta) + \cos(2\Re \theta)} < 1.$$

The denominator is positive and the branch satisfies $\arctan(\tan \theta) = \theta$, first on a neighbourhood of zero and then on this connected strip by analytic uniqueness. Hence (GC.25)–(GC.26) extend to that strip with the analytic branch of the cosine power in (GC.13). No logarithm of $\mathcal{A}_h$ is asserted: it can have complex zeros, whereas all real-tilt integrals $M_h(\theta)$ are strictly positive.

For each finite derivative there is also an exact moment formula. Write $b_n^{(\lambda)}(t) = \sum_{j=0}^n b_{n,j} t^j$. Then

$$n!(\alpha)_n c_{h,n} = \frac{1}{c_\lambda} \sum_{j=0}^n b_{n,j} M_h^{(j)}(0), \quad d_{h,k,n} = \frac{1}{n!} \left. \frac{d^n}{dz^n} \mathcal{A}_h(z)^k \right|_{z=0}. \quad (\text{GC.30})$$

The first identity follows by the finite polynomial expansion and (GC.9); the second is Taylor's formula, whose Leibniz sum is exactly (GC.22). The triangular coefficients $b_{n,n} = 1$ give an invertible finite map between moments through degree n and coefficients through degree n , with the full diagonal factors $c_\lambda n!(\alpha)_n$ retained.

5 The complete density expansion for the original bounded references

The finite identities and analytic join above hold for every $\lambda > 0$. For the original full-quartet reference choose $\lambda = 1/4$. More generally choose the actual shifted reference

$$\lambda = \frac{1}{4} + L, \quad L \text{ an integer}, \quad L \geq \max(0, 3 - d). \quad (\text{GC.31})$$

These choices admit the following proved bounded multiplier and hence the complete L^2 density expansion.

Indeed at $s = 1/2 + it$ the defining g identity gives

$$A_{h,1/4}(t) = \frac{-\pi^{-1/4}(t^2 + 1/4)e^{-it \log \pi/2} \zeta(1/2 + it)}{h(1/2 + it)}. \quad (\text{GC.32})$$

This expression is continued through each selected zero by the already defined entire quotient. Integrating the step function $[x]$, initially for $\Re s > 1$, gives $\zeta(s) = s/(s-1) - s \int_1^\infty \{x\} x^{-s-1} dx$. The last integral is holomorphic for $\Re s > 0$, so the identity continues there away from 1. On the original line it gives $|\zeta(s)| \leq 1 + 2\sqrt{t^2 + 1/4} \leq 2(1 + |t|)$. For a nonempty packet put $T_h = \max(1, 2 \max_\rho |\rho - 1/2|)$; for $h = 1$ put $T_h = 1$. For $|t| \geq T_h$, every original packet factor has modulus at least $|t|/2$. Gamma recurrence then gives

$$\begin{aligned} A_{h,1/4+L}(t) &= \frac{A_{h,1/4}(t)}{\prod_{j=0}^{L-1} (j + 1/4 + it/2)}, \\ B_{h,1/4+L}(t) &\leq \frac{25 \cdot 2^{2(d+L)}}{\sqrt{\pi}} |t|^{6-2d-2L}. \end{aligned} \quad (\text{GC.33})$$

For the constant 25, the two squared numerator estimates at $|t| \geq 1$ are $(t^2 + 1/4)^2 \leq 25t^4/16$ and $|\zeta(s)|^2 \leq 16t^2$. Each shift factor has modulus at least $|t|/2$. The exponent in (GC.33) is nonpositive for (GC.31). Thus a completely specified finite bound is

$$C_{h,L} = \max \left\{ \sup_{|t| \leq T_h} B_{h,1/4+L}(t), \frac{25 \cdot 2^{2(d+L)}}{\sqrt{\pi}} T_h^{6-2d-2L} \right\} > 0. \quad (\text{GC.34})$$

The compact term is finite by continuity of the cancelled multiplier. This is a mathematical bound with a specified supremum, not a claim that a numerical packet supremum has already been enclosed.

For these references $B \in L^2(d\sigma_\lambda)$, because the reference has finite mass and $0 \leq B \leq C_{h,L}$. Fibre integration of $w_h = Br_\lambda$ proves

$$B_{h,\lambda;k} = C_\Sigma(B^{\otimes k}), \quad m_{h,k} = r_{\lambda,k} B_{h,\lambda;k}, \quad 0 \leq B_{h,\lambda;k} \leq C_{h,L}^k. \quad (\text{GC.35})$$

For $k \geq 2$ it is positive at every sum coordinate: the zeros of B form a countable discrete set, and their forbidden fibre coordinates are a countable union of affine hyperplanes of measure zero. Every remaining reference factor is positive. For $k = 1$ the original isolated zeros remain.

Combining (GC.23), completeness, and the sum norm in (GC.15) gives the full, convergent $L^2(r_{\lambda,k} du)$ identity

$$B_{h,\lambda;k}(u) = \sum_{n \geq 0} \frac{d_{h,k,n}}{(k\alpha)_n} b_n^{(k\lambda)}(u). \quad (\text{GC.36})$$

In particular the norm of the retained relative density vector is

$$\left\| B^{\otimes k} - U_{\Sigma} B_{h,\lambda;k} \right\|^2 = \left(\int B^2 d\sigma_{\lambda} \right)^k - c_{\lambda}^k \sum_{n \geq 0} \frac{n!}{(k\alpha)_n} |d_{h,k,n}|^2. \quad (\text{GC.37})$$

This is Pythagoras for (GC.17), so every norm and mass factor is retained. A truncation of (GC.36) is not asserted pointwise positive.

6 Original-coordinate finite Grams and every tilt derivative

Let $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$ for $N \geq 0$, with original ordered basis $1, S, \dots, S^N$, and set $c = k/2$. The arithmetic source Gram and its reference are

$$\begin{aligned} (\mathbf{M}_{h,k,N}(\theta))_{ij} &= \int (c - iu)^i (c + iu)^j e^{\theta u} m_{h,k}(u) du, \\ (\mathbf{M}_{\lambda,k,N}^{\Gamma}(\theta))_{ij} &= \int (c - iu)^i (c + iu)^j e^{\theta u} r_{\lambda,k}(u) du, \quad 0 \leq i, j \leq N. \end{aligned} \quad (\text{GC.38})$$

They are holomorphic entrywise for $|\Re \theta| < \pi/2$ and positive definite at every real tilt in that strip: a nonzero polynomial has only finitely many real-line roots and each measure is positive almost everywhere. Let $\mathbf{H}_{h,k,N}$ denote the corresponding real-u monomial Hankel matrix. The exact matrix of the unchanged coordinate substitution is

$$(\mathbf{L}_{N+1})_{aj} = \begin{cases} \binom{j}{a} c^{j-a} i^a, & 0 \leq a \leq j \leq N, \\ 0, & a > j, \end{cases} \quad \det \mathbf{L}_{N+1} = i^{N(N+1)/2}. \quad (\text{GC.39})$$

Expanding each original S monomial proves

$$\mathbf{M}_{h,k,N} = \mathbf{L}_{N+1}^* \mathbf{H}_{h,k,N} \mathbf{L}_{N+1} \quad (\text{GC.40})$$

and the identical congruence for the reference. Since the displayed determinant has modulus one, it also proves exact equality of the two coordinate determinants. The congruence, rather than a relabeling of a matrix, preserves every complex phase in its entries.

For an integer $r \geq 0$, expand the specified polynomial uniquely as

$$u^r (c - iu)^i (c + iu)^j = \sum_{n=0}^{i+j+r} e_{ijr,n}^{(k,\lambda)} b_n^{(k\lambda)}(u). \quad (\text{GC.41})$$

The monic triangular basis makes these coefficients finite, unique, and computable by polynomial subtraction in the written order. Differentiation of (GC.38) and (GC.23) now give the exact formula

$$\boxed{\partial_{\theta}^r (\mathbf{M}_{h,k,N})_{ij} |_{\theta=0} = c_{\lambda}^k \sum_{n=0}^{i+j+r} e_{ijr,n}^{(k,\lambda)} n! d_{h,k,n}.} \quad (\text{GC.42})$$

The derivative interchange is justified by the exponential bounds after (GC.9), or directly by Fubini on the k-fold product. Since $d_{h,k,n}$ uses exactly the finite coefficient power (GC.22), the entire rth derivative matrix is determined by $c_{h,0}, \dots, c_{h,2N+r}$. At zero tilt $r = 0$, the list ends at $2N$. There is no truncation error in this finite statement.

Equation (GC.42) supplies each derivative at zero from its stated finite coefficient list. Evaluation at a nonzero tilt is related to those derivatives by the following exact analytic Taylor identity, valid for $|\theta| < \pi/2$:

$$(M_{h,k,N}(\theta))_{ij} = \sum_{r \geq 0} \frac{\theta^r}{r!} c_\lambda^k \sum_{n=0}^{i+j+r} e_{ijr,n}^{(k,\lambda)} n! d_{h,k,n}. \quad (\text{GC.43})$$

Absolute convergence follows by bounding the absolute exponential series by $e^{|\theta||u|}$ times the original polynomial modulus. For larger imaginary components within the full holomorphic strip, analytic continuation of these entries is the defined original Laplace integral (GC.38). This specifies the relation between all finite coefficient jets and the full tilted metric.

7 The original cyclic quotient and the exact cochain correction

For a nonempty packet let $E_h = \mathbb{C}[s]/(h)$ and $\mathcal{E}_h = E_h^{\otimes k}$. Let a be the element $s_1 + \dots + s_k$ of this finite algebra. The kernel of $\mathbb{C}[S] \rightarrow \mathcal{E}_h$, $P \mapsto P(a)$, is generated by its unique monic polynomial χ . It is nonzero since the powers of a lie in a finite-dimensional vector space. Put $q = \deg \chi \geq 1$, $C = \mathbb{C}[S]/(\chi)$, and $A = M_S : C \rightarrow C$. The induced map $C \rightarrow \mathcal{E}_h$ is injective by the definition of the kernel. For each selected zero, the complete Taylor value of g/h is nonzero at its centre; its residue $v_h = j_h(g/h) \in E_h$ is consequently a unit in every local factor and in E_h . Thus the original arithmetic injection

$$\eta_{h,k} : C \hookrightarrow \mathcal{E}_h, \quad [P] \mapsto v_h^{\otimes k} P(a) \quad (\text{GC.44})$$

retains the entire local unit and every selected multiplicity.

The original analytic polynomial map is

$$\mathcal{V}_{h,k} P = P(D_1 + \dots + D_k) F_h^{\otimes k}. \quad (\text{GC.45})$$

Its tensor Mellin transform is $P(s_1 + \dots + s_k) \prod_i v_h(s_i)$. Taking complete selected jets gives precisely (GC.44) after the quotient π_χ . No modulus operation has been applied to that arithmetic jet map.

Successive monic divisions, in the order $s_1, \dots, s_k$, provide specific polynomials $Q_i(s_1, \dots, s_k)$ with

$$\chi(s_1 + \dots + s_k) = \sum_{i=1}^k h(s_i) Q_i(s_1, \dots, s_k). \quad (\text{GC.46})$$

To prove this, divide in s_1 , retain its remainder, divide that remainder in s_2 , and continue. The final remainder has degree less than d in every coordinate. Its residue in $\mathcal{E}_h$ is zero by the defining property of χ , and the tensor remainder monomials form a basis, so every coefficient is zero. This proves the displayed identity and fixes the division order.

The operators D_i commute on the tensor complex. For a polynomial P , define the actual cochain of degree $k-1$ by

$$\begin{aligned} \mathcal{P}_{\chi,h,k}(P) = \sum_{i=1}^k (-1)^{i-1} Q_i(D_1, \dots, D_k) P(D_1 + \dots + D_k) \\ \cdot (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \end{aligned} \quad (\text{GC.47})$$

Every summand belongs to the indicated tensor source, since D preserves V and $\mathcal{B}$. The first $i-1$ factors have cochain degree one. The tensor differential on the i th factor contributes $(-1)^{i-1}$, cancelling the written sign. The other factors have zero differential. Equations (GC.8) and (GC.46) therefore give

$$d_{\text{tot}} \mathcal{P}_{\chi,h,k}(P) = \mathcal{V}_{h,k}(\chi P). \quad (\text{GC.48})$$

This identifies the cochain primitive, its sign, and its boundary in their different cochain degrees.

For the full quotient at finite source degree take $N \geq q - 1$ and $m = N - q + 1$. With $\mathcal{P}_{-1} = 0$, monic division gives the exact sequence in original coordinates

$$0 \longrightarrow \mathcal{P}_{m-1} \xrightarrow{M_\chi} \mathcal{P}_N \xrightarrow{J_N} C \longrightarrow 0, \quad M_\chi P = \chi P, \quad J_N P = [P]_\chi. \quad (\text{GC.49})$$

The kernel assertion is divisibility, and the first q monomial columns prove surjectivity. Let B_χ be the $(N+1) \times m$ matrix of M_χ in the original ordered bases. Its entries are $[S^i](\chi(S)S^j)$; it is independent of the metric and tilt. The relation Gram and every zero-tilt derivative are therefore

$$B_{h,k,N}(\theta) = B_\chi^* M_{h,k,N}(\theta) B_\chi, \quad B_{h,k,N}^{(r)}(0) = B_\chi^* M_{h,k,N}^{(r)}(0) B_\chi. \quad (\text{GC.50})$$

Equivalently its polynomial integrand is $u^r |\chi(c+iu)|^2 (c-iu)^i (c+iu)^j$, where $0 \leq i, j < m$. Its degree is at most $r + 2q + 2(m-1) = r + 2N$, so the relation derivative uses the same list through $2N + r$ as the source derivative. Both complex polynomial factors, their conjugates, and all χ coefficients remain in this formula.

For a fixed real admitted tilt write $M_h = M_{h,k,N}(\theta)$ and $M_\Gamma = M_{\lambda,k,N}^\Gamma(\theta)$. Their canonical quotient Grams and minimum lifts are

$$G_h = (J_N M_h^{-1} J_N^*)^{-1}, \quad R_h = M_h^{-1} J_N^* G_h, \quad G_\Gamma = (J_N M_\Gamma^{-1} J_N^*)^{-1}, \quad R_\Gamma = M_\Gamma^{-1} J_N^* G_\Gamma. \quad (\text{GC.51})$$

Surjectivity of J and positivity of the source Grams prove every inverse exists. These formulas give $JR_h = JR_\Gamma = I_C$ and make each lift orthogonal to $\ker J$ in its own metric, proving its minimum property by the orthogonal Pythagorean identity. For $m > 0$ put

$$K = (B_\chi^* M_h B_\chi)^{-1} B_\chi^* M_h R_\Gamma : C \longrightarrow \mathbb{C}^m. \quad (\text{GC.52})$$

The relation Gram is positive definite because B_χ is injective. The section $R_\Gamma - B_\chi K$ still has quotient identity and is orthogonal to the relation space in M_h . The preceding uniqueness gives the exact section map

$$\boxed{R_h = R_\Gamma - B_\chi K.} \quad (\text{GC.53})$$

At $N = q - 1$, $m = 0$, take K to be the zero map to the zero vector space and omit the inverse in (GC.52); (GC.53) remains the identity of the unique degree-less-than- q representatives.

For $u_0 \in C$, let $\kappa_{u_0}(S)$ be the polynomial whose coefficient vector is Ku_0 . Combining (GC.48) and (GC.53) proves the exact cochain correction

$$\boxed{\mathcal{V}_{h,k}(R_h u_0 - R_\Gamma u_0) = -\mathcal{V}_{h,k}(\chi \kappa_{u_0}) = d_{\text{tot}}(-\mathcal{P}_{\chi,h,k}(\kappa_{u_0})).} \quad (\text{GC.54})$$

Thus applying $\mathcal{V}$ gives the boundary in degree k ; the explicit negative cochain on the right is its primitive in degree $k - 1$. Passing to the original arithmetic quotient sends this boundary to zero. Explicitly, for the original set of labels $\mathcal{L}$ and vector spaces E_ℓ , write the supplied carrier as $\{\tau\} \sqcup \bigsqcup_{\ell \in \mathcal{L}} (\{\ell\} \times E_\ell)$. A family of linear maps $L_\ell : E_\ell \rightarrow F_\ell$ has the exact support lift $\tau \mapsto \tau$ and $(\ell, v) \mapsto (\ell, L_\ell v)$, with $e_\ell = (\ell, 0)$ in each target fibre. Substitution proves that these lifts preserve identities and composition. Applied to the original quotient, the relation boundary in (GC.54) therefore has image e_ℓ , while the external τ has image τ . The source vector and its primitive remain available before that quotient.

8 Determinant indices and the original Toda substitution

For real admitted θ , write $\mathfrak{D}_n$ for the original degree-less-than-n source Gram determinant and $\mathfrak{B}_m$ for the relation Gram determinant on m multiplication columns. Empty determinants are one. The analogous reference determinants carry the superscript Γ . For $N \geq q - 1$, set

$$V_N = \det G_h, \quad V_N^\Gamma = \det G_\Gamma, \quad X_n = \mathfrak{D}_n / \mathfrak{D}_n^\Gamma, \quad Y_m = \mathfrak{B}_m / \mathfrak{B}_m^\Gamma.$$

The quotient-first coefficient matrix with columns $1, S, \dots, S^{q-1}, \chi, \chi S, \dots, \chi S^{m-1}$ has determinant one: its columns have successively distinct degrees and all their leading coefficients are one. Replacing its first q columns by the corresponding minimum sections only subtracts relation columns, so it still has determinant one. Orthogonality then gives the exact block determinant formula

$$V_N = \frac{\mathfrak{D}_{N+1}}{\mathfrak{B}_{N-q+1}} = V_N^\Gamma T_N, \quad T_N = \frac{X_{N+1}}{Y_{N-q+1}}, \quad N \geq q - 1. \quad (\text{GC.55})$$

In particular the first full quotient has denominator $\mathfrak{B}_0 = 1$ and requires no relation inverse.

For completeness the reference source determinant is

$$\mathfrak{D}_n^\Gamma(\theta) = c_\lambda^{kn} \prod_{j=0}^{n-1} j!(k\alpha)_j (\cos \theta)^{-n(k\alpha+n-1)}, \quad a_n^\Gamma(\theta) = n(n + k\alpha - 1) \sec^2 \theta. \quad (\text{GC.56})$$

Here a_n is the ratio of consecutive monic squared norms in the real-u coordinate; the norm-preserving phase of (GC.39) gives the same ratios in the original S coordinate. A direct proof of the Hankel identity used to obtain (GC.56) is as follows. Let p_j be the monic orthogonal polynomials for an $e^{\theta u}$ -tilted positive measure, let h_j be their squared norms, and write $up_j = p_{j+1} + b_j p_j + a_j p_{j-1}$, with $p_{-1} = 0$ and $a_0 = 0$. For $j \geq 1$, pairing with p_{j-1} gives $a_j = h_j / h_{j-1}$, and pairing with p_j gives $h'_j = b_j h_j$ for every $j \geq 0$. Differentiating orthogonality for $j \geq 1$ shows $\partial_\theta p_j = -a_j p_{j-1}$: its pairings with degrees at most $j - 2$ vanish, and the remaining pairing is $-h_j$. For $j \geq 1$, write $p_j = u^j + c_j u^{j-1} + \dots$, and set $c_0 = 0$ for $p_0 = 1$. The derivative statement for p_0 is $\partial_\theta p_0 = 0$. Thus $c'_j = -a_j$ and $b_j = c_j - c_{j+1}$ hold for every $j \geq 0$, and consequently $b'_j = a_{j+1} - a_j$. Since $\mathfrak{D}_n = \prod_{j < n} h_j$,

$$(\log \mathfrak{D}_n)'' = \sum_{j < n} b'_j = a_n = \mathfrak{D}_{n+1} \mathfrak{D}_{n-1} / \mathfrak{D}_n^2, \quad n \geq 1.$$

For the reference, the determinant values at $n=0$ and $n=1$ are 1 and (GC.13). The right side of the proposed formula (GC.56) has second logarithmic derivative $n(k\alpha + n - 1) \sec^2 \theta$. Its consecutive ratio has exactly the same value, with constants $n!(k\alpha)_n / ((n-1)!(k\alpha)_{n-1})$ and cosine exponent -2 . Positive induction in the displayed determinant identity proves (GC.56) for every n .

The reference relation determinant is the determinant of the moment matrix with entries

$$\partial_\theta^{i+j} \{ \overline{\chi}(c - i\partial_\theta) \chi(c + i\partial_\theta) Z_{\lambda,k}^\Gamma(\theta) \}, \quad 0 \leq i, j < m. \quad (\text{GC.57})$$

Differentiation under the integral gives the unchanged weight $|\chi(c + iu)|^2 r_{\lambda,k}(u)$ in this u-coordinate matrix. The relation version of (GC.39)–(GC.40) has determinant of modulus one, proving that (GC.57) is the original S-coordinate relation determinant as stated. It contains the actual χ with all coefficients.

At $\theta = 0$ put

$$Q_N = \frac{X_{N+2} X_N}{X_{N+1}^2}, \quad \mathcal{R}_N^\Gamma = \frac{V_{N-1}^\Gamma}{V_{N+1}^\Gamma}.$$

The consecutive full-quotient formulas have domain $N \geq q$ and give, by literal substitution in (GC.55)–(GC.56),

$$a_{N+1} = (N+1)(N+k\alpha)Q_N, \quad \frac{V_{N-1}}{V_{N+1}} = \mathcal{R}_N^\Gamma \frac{T_{N-1}}{T_{N+1}}. \quad (\text{GC.58})$$

The domain $N \geq q$ is necessary because V_{N-1} occurs. The separate original endpoint at $N = q - 1$ is not replaced by a q -dimensional metric at degree $q - 2$. At every fixed admissible real tilt the phase derivative remains the exact equality

$$\partial_\theta \log V_N = \partial_\theta \log V_N^\Gamma + \partial_\theta \log T_N, \quad N \geq q - 1. \quad (\text{GC.59})$$

All these derivatives are of positive real-tilt determinants. Formula (GC.42) gives their underlying source and relation entry derivatives with the precise finite coefficient indices; their quotient and determinant operations use those original matrices.

For $h = 1$ the finite arithmetic quotient is zero and $\chi = 1$. Its relation and source spaces coincide at every degree N , so $\mathfrak{B}_{N+1} = \mathfrak{D}_{N+1}$, $V_N = V_N^\Gamma = T_N = 1$. The quotient lift has zero-dimensional domain and the cochain correction (GC.54) is the zero map on that domain. The nonzero analytic density, its mass μ_1^k , its coefficient series and (GC.24)–(GC.43) remain as constructed. The support lift is $G(0) = \{\tau, e\}$; it does not produce a nonzero finite spectral quotient from the analytic seed.